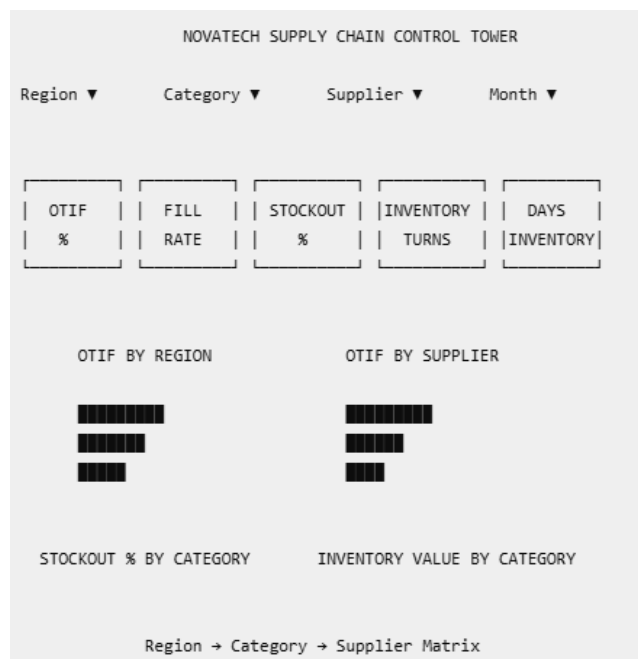


The objective is for students to go through:

CSV → Power Query → Data Model → DAX Measures → KPI Cards → Drill-down → Management Insight

Use the CSV version here: [NovaTech_Supply_Chain_Analytics_Dataset.csv](#)



1. Start with a management challenge

COO Challenge: “I don’t want to see 50 metrics. Build me a one-page dashboard showing whether we are serving customers well while controlling inventory.”

We will create following 5 metric: **OTIF | Fill Rate | Stockout % | Inventory Turns | Days Inventory**

Then they must answer: **“Where is the problem, and what should management investigate?”**

2. Import the NovaTech Dataset

In Power BI Desktop:

Home → Get Data → Text/CSV → NovaTech_Supply_Chain_Analytics_Dataset.csv

Then choose: **Transform Data**

Check these data types:

Column	Type
Date	Date
Demand	Whole Number
Forecast	Whole Number
Opening Inventory	Whole Number
Closing Inventory	Whole Number
Lead Time (Days)	Whole Number
Order Qty	Whole Number
Delivered Qty	Whole Number
Requested Date	Date
Delivery Date	Date
Unit Cost (INR)	Decimal
Inventory Value (INR)	Decimal
OTIF	Text
Stockout	Text

Then: **Close & Apply**

3. Create a Measures Table

Go to: **Home** → **Enter Data**

Create a dummy table called: KPI Measures

Then store all your DAX measures there.

4. KPI 1 — OTIF

First explain the business logic. An order qualifies as OTIF only when it is: **On Time AND In Full**

Your generated dataset already contains the OTIF column with Yes/No, so students can create:

Total Orders = COUNTROWS('Supply Chain Data')
OTIF Orders = CALCULATE(COUNTROWS('Supply Chain Data'), 'Supply Chain Data'[OTIF] = "Yes")
OTIF % = DIVIDE([OTIF Orders], [Total Orders], 0)

Format OTIF % as percentage.

Put it on a Card

Visualizations → **Card**

Field: OTIF %

Question: "Is this number good or bad?" Let's slice it.

5. Make OTIF analytical

Add three slicers:

- **Region**
- **Category**
- **Supplier**

Now click: North, and compare OTIF. Then: Critical Components, Then suppliers.

Aggregate → **Segment** → **Drill Down** → **Root Cause**

6. KPI 2 — Fill Rate

- OTIF asks whether the **order** was delivered correctly.
Fill Rate asks how much of the **quantity requested** was actually fulfilled.

Create:

Total Ordered Qty = SUM('Supply Chain Data'[Order Qty])
Total Delivered Qty = SUM('Supply Chain Data'[Delivered Qty])

```

Fill Rate % =
DIVIDE(
    [Total Delivered Qty],
    [Total Ordered Qty],
    0
)

```

Put Fill Rate % on another Card.

Interview question: **“Can Fill Rate be 98% while OTIF is only 89%?”** Answer: Yes.

Orders might eventually be fulfilled, but some could have been late or not delivered in full at the required time. This distinction is important.

7. KPI 3 — Stockout %

Our dataset has: Stockout = Yes/No

```

Stockout Events =
CALCULATE(
    COUNTROWS('Supply Chain Data'),
    'Supply Chain Data'[Stockout] = "Yes"
)

```

```

Stockout % =
DIVIDE(
    [Stockout Events],
    [Total Orders],
    0
)

```

For this dataset, each row represents a demand/order event, so Total Orders is serving as the denominator for demand events.

Now add a bar chart:

- **Axis:** Category
- **Value:** Stockout %

Students should immediately see differences between categories.

Then change: **Axis → Supplier**

Then: **Axis → Region**

Now Power BI becomes diagnostic analytics.

8. KPI 4 — Average Inventory

Before Inventory Turns, calculate average inventory.

At row level conceptually:

Average Inventory = (Opening Inventory + Closing Inventory) / 2

A simple measure is:

```

Average Inventory Units =
DIVIDE(
    SUM('Supply Chain Data'[Opening Inventory]) +
    SUM('Supply Chain Data'[Closing Inventory]),
    2
)

```

But inventory turns should normally be calculated on a **value basis**, not by mixing quantities across SKUs with different costs.

Therefore create:

```
Average Inventory Value =  
SUMX(  
    'Supply Chain Data',  
    DIVIDE(  
        'Supply Chain Data'[Opening Inventory] +  
        'Supply Chain Data'[Closing Inventory],  
        2  
    )  
    *  
    'Supply Chain Data'[Unit Cost (INR)]  
)
```

This is also a nice opportunity to explain SUM versus SUMX.

9. KPI 5 — COGS

Our training dataset doesn't contain an explicit COGS column. We can approximate product cost associated with delivered quantity as:

```
COGS =  
SUMX(  
    'Supply Chain Data',  
    'Supply Chain Data'[Delivered Qty] *  
    'Supply Chain Data'[Unit Cost (INR)]  
)
```

In a real ERP/finance model, COGS should come from the appropriate accounting/transaction data rather than being casually reconstructed.

10. KPI 6 — Inventory Turns

```
Inventory Turns =  
DIVIDE(  
    [COGS],  
    [Average Inventory Value],  
    0  
)
```

11. KPI 7 — Days Inventory

If you have annualized inventory turns:

```
Days Inventory =  
DIVIDE(  
    365,  
    [Inventory Turns],  
    0  
)
```

Example: Inventory Turns = 5 Therefore: **Days Inventory = 73 days** “Would lower DOI always be better?”

The objective isn't minimum inventory. It's the **right inventory for the required service level**.

12. Build the first Power BI page

COO SUPPLY CHAIN CONTROL TOWER

Top row:

OTIF	Fill Rate	Stockout	Inventory Turns	Days Inventory
------	-----------	----------	-----------------	----------------

Use five Card visuals. Underneath, create four charts.

OTIF by Region

- Clustered bar chart:
- **Axis:** Region
Value: OTIF %

OTIF by Supplier

- **Axis:** Supplier
Value: OTIF %
- Sort ascending.
- Immediately the worst suppliers become visible.

Stockout % by Category

- **Axis:** Category
Value: Stockout %

Inventory Value by Category

- **Axis:** Category
Value:

```
Inventory Value =  
SUM('Supply Chain Data'[Inventory Value (INR)])
```

Now we have a basic control tower.

13. Add a KPI that connects Supply Chain to Finance

Create:

```
Inventory Value =  
SUM('Supply Chain Data'[Inventory Value (INR)])
```

- Put it on a Card.
- Format it as ₹.

Now ask: “Which categories are consuming our working capital?”

Create a Treemap:

- **Group:** Category
- **Values:** Inventory Value

Then drill into:

Category → Product → SKU

This makes the financial consequence of inventory visible.

14. Create the most important diagnostic visual

Create a **Matrix**.

Rows: Region, then Category, then Supplier

Values:

- OTIF %

- Fill Rate %
- Stockout %
- Inventory Value
- Lead Time

For lead time:

```
Average Lead Time =
AVERAGE('Supply Chain Data'[Lead Time (Days)])
```

We can now expand: **Region → Category → Supplier**

This is where they can discover: Company OTIF is poor → Which region? → Which category? → Which supplier?

That's real diagnostic analytics.

15. Turn the exercise into a 5-minute investigation

Tell the COO:

1. What is our OTIF?
2. Which region has the biggest service problem?
3. Which product category needs attention?
4. Which supplier should management investigate?
5. Where is inventory/working capital concentrated?

Then: *Recommend ONE management action.*

16. Introduce the OTIF trap

After they have used the existing OTIF field, teach them how OTIF should actually be constructed.

Create:

```
On Time Orders =
CALCULATE(
    COUNTROWS('Supply Chain Data'),
    'Supply Chain Data'[Delivery Date]
        <= 'Supply Chain Data'[Requested Date]
)
```

In-full requires:

```
In Full Orders =
CALCULATE(
    COUNTROWS('Supply Chain Data'),
    'Supply Chain Data'[Delivered Qty]
        >= 'Supply Chain Data'[Order Qty]
)
```

The Trap

Suppose:

1,000 orders

920 On-Time

900 In-Full

but only: **870 orders satisfy BOTH conditions**

Then:

On-Time = 92%

In-Full = 90%

but: **OTIF = 87%**

OTIF is the **intersection**. You can demonstrate this directly with DAX:

```
Calculated OTIF Orders =  
CALCULATE(  
    COUNTROWS('Supply Chain Data'),  
    FILTER(  
        'Supply Chain Data',  
        'Supply Chain Data'[Delivery Date]  
            <= 'Supply Chain Data'[Requested Date]  
        &&  
        'Supply Chain Data'[Delivered Qty]  
            >= 'Supply Chain Data'[Order Qty]  
    )  
)
```

Then:

```
Calculated OTIF % =  
DIVIDE(  
    [Calculated OTIF Orders],  
    [Total Orders],  
    0  
)
```

Now compare Calculated OTIF % with the supplied OTIF column.

* * *